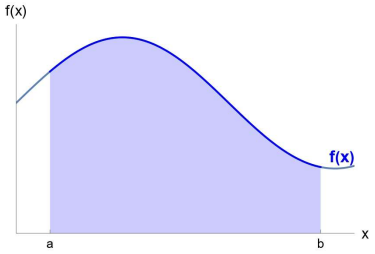
6.3 Introduction to Double Integration

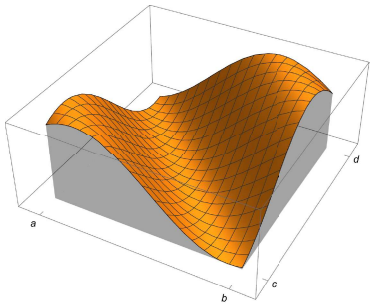
As we saw in Chapter 3, probability density functions often serve as the continuous analogue of probability mass functions. To compute probabilities from a density function, we integrate over an interval. Extending this idea to jointly continuous random variables requires evaluating integrals of functions $f : \mathbb{R}^2 \to \mathbb{R}$, which involves integration over two variables-this is known as double integration.

In Semester 2, the course Multivariable Calculus will introduce double integration. Here, we provide a brief overview to support its use in probability theory. The purpose of this section is not to cover the material in that course but to merely introduce you to key concepts that we can use for joint distributions.

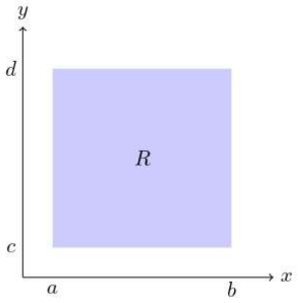
One way to understand integration is as the calculation of the area under a curve. We extend this idea to functions of two variables by computing the volume under a surface. For example, the integral of the indicator function $\mathbb{I}_{[a,b]}$ gives the length of the interval, $b-a$, while the double integral of the indicator function $\mathbb{I}_{[a,b]\times[c,d]}$ yields the area of the rectangle, $(b-a)(d-c)$.



(a) $\int_a^b f(x)\,\mathrm{d}x$



(b) $\int_a^b \int_c^d f(x,y)\,\mathrm{d}x\,\mathrm{d}y$



(c) $R = [a,b] \times [c,d]$

Double integrals such as

$$\iint_R f(x,y)\,\mathrm{d}x\;\mathrm{d}y$$

consist of three components: i) the integrand f; ii) the region of integration R; iii) the order of integration (e.g. $\mathrm{d}x\,\mathrm{d}y$). We can calculate double integrals by reducing them to repeated integrals. We discuss how to do this in a few cases.

Rectangular Domains

The simplest situation to consider is that of rectangular domains, i.e. when the domain of integration is a rectangular region $R = \{(x,y) | a \le x \le b, c \le y \le d\}$. In this case we can write the double integral as the repeated integral

$$\iint_R f(x,y)\,\mathrm{d}x\;\mathrm{d}y = \int_c^d \int_a^b f(x,y)\,\mathrm{d}x\;\mathrm{d}y = \int_c^d \left(\int_a^b f(x,y)\,\mathrm{d}x \right) \mathrm{d}y.$$

If we inspect the inner integral we note that y is fixed and the integral is only happening with respect to the variable x. As such we can just treat y as a constant when calculating the inner integral and evaluate

$$\int_a^b f(x,y)\,\mathrm{d}x$$

as a typical one-dimensional integral. Once we have evaluated this integral we have a function of only one variable y (which also depends on a and b) and we need to integrate this from c to d which is again a one dimensional integral. In this way we have reduced the two dimensional integration into 2 one-dimensional integrals.

Example 6.5. Consider

$$\int_0^1 \int_1^2 xy \, \mathrm{d}x \, \mathrm{d}y.$$

The first step is to plot the area of integration; this is shown in Figure 6.2. After plotting the

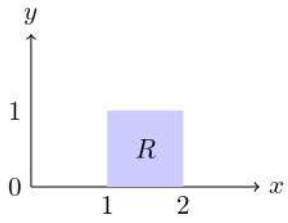


Figure 6.2: The region $R = [1, 2] \times [0, 1]$

area of integration our next step is to evaluate the inner integral, that is, we calculate

$$\int_1^2 xy \, \mathrm{d}x = \left[\frac{1}{2}x^2 y\right]_{x=1}^{x=2} = \frac{1}{2}4y - \frac{1}{2}y = \frac{3}{2}y.$$

Now to calculate the double integral we have

$$\int_0^1 \int_1^2 xy \, \mathrm{d}x \, \mathrm{d}y = \frac{3}{2}\int_0^1 y \, \mathrm{d}y = \frac{3}{2}\left[\frac{1}{2}y^2\right]_{y=0}^{y=1} = \frac{3}{4}.$$

Remark 6.1 (A comment on notation). Sometimes we write the integral as

$$\iint_R f(x, y) \, \mathrm{d}x \, \mathrm{d}y \text{ or } \int_c^d \int_{a(y)}^{b(y)} f(x, y) \, \mathrm{d}x \, \mathrm{d}y.$$

Here R is the two dimensional region $\{(x, y)|a(y) \leq x \leq b(y), c \leq y \leq d\}$. In the case of rectangular domains a and b are constants. In Example 6.5 the region of integration is shown in Figure 6.2. Under general assumptions (which will hold for all the examples in this course) you can swap the order of integration and do the y integral first followed by the x integral. Always keep track of the limits when you swap the order! For more details see Multivariable Calculus, Fubini's Theorem.

Example 6.6 (Example 6.5 continued)**.** We have already seen that

$$\int_0^1 \int_1^2 xy \, \mathrm{d}x \, \mathrm{d}y = \frac{3}{4}.$$

Let us verify for this example that we can swap the order of integration, i.e.

$$\int_0^1 \int_1^2 xy \, \mathrm{d}x \, \mathrm{d}y = \int_1^2 \int_0^1 xy \, \mathrm{d}y \, \mathrm{d}x.$$

We first calculate the inner integral

$$\int_0^1 xy\,\mathrm{d}y = \left[\frac{1}{2}xy^2\right]_{y=0}^{y=1} = \frac{1}{2}x.$$

To calculate the double integral we have

$$\int_0^1 \int_1^2 xy\,\mathrm{d}x\ \mathrm{d}y = \int_1^2 \frac{1}{2}x\,\mathrm{d}x = \left[\frac{1}{4}x^2\right]_{x=1}^{x=2} = \frac{3}{4}.$$

Triangular Domains

The next type of domain we consider are triangular domains, which is when the region of integration R is a triangular shape. For example

$R = \{(x, y)|0 \leq x \leq y, 0 \leq y \leq 1\}$.

A good first step for any double integration question is to draw the domain. For this example a sketch of R is given in Figure 6.3. A useful approach to sketching R is to first plot the boundary lines, then test a few points to identify the relevant region.

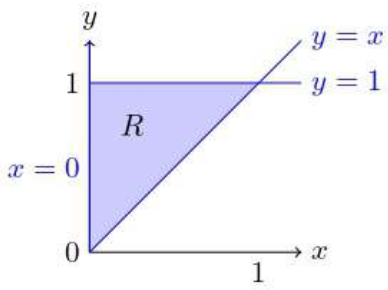


Figure 6.3

As with rectangular domains, we can evaluate double integrals over triangular domains by evaluating repeated integrals, only now we need to be aware that the limits of the integral are also functions of the variable.

Example 6.7. Consider how to calculate

$$\iint_R xy\,\mathrm{d}x\,\mathrm{d}y$$

where $R = \{(x, y)|0 \leq x \leq y, 0 \leq y \leq 1\}$. Our first step is to draw the region which we have already done in Figure 6.3. In order to write this as a repeated integral we want to write it in the form

$$\int_c^d \int_{a(y)}^{b(y)} f(x, y)\,\mathrm{d}x\ \mathrm{d}y.$$

From the figure we see that y can vary between 0 and 1 then for each fixed y the variable x takes values from 0 to y, so we have $c = 0, d = 1, a(y) = 0$ and $b(y) = y$, that is

$$\iint_R xy\,\mathrm{d}x\,\mathrm{d}y = \int_0^1 \int_0^y xy\,\mathrm{d}x\,\mathrm{d}y.$$

Note that we could have chosen to do the integral in the other order, in which case x is the variable of the outermost integral. In this case x takes values from 0 to 1, then for each fixed x the variable y varies between y to 1. Therefore we have

$$\iint_R xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \int_0^y xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \int_x^1 xy \,\mathrm{d}y \,\mathrm{d}x$$

We can use either of these to evaluate the integral

$$\iint_R xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \int_0^y xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 y \left[\frac{1}{2}x^2\right]_{x=0}^{x=y} \mathrm{d}y = \int_0^1 \frac{1}{2}y^3 \,\mathrm{d}x = \left[\frac{1}{8}y^4\right]_{y=0}^{y=1} = \frac{1}{8}.$$

Note in the examples we have considered so far there has been little reason to change the order of integration. However sometimes it can be very important, consider for example

$$\int_0^1 \int_y^1 e^{-x^2} \,\mathrm{d}x \,\mathrm{d}y.$$

In this case there is no explicit formula for the integral of e^{-x^2} however if we exchange the integrals we have

$$\int_0^1 \int_y^1 e^{-x^2} \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \int_0^x e^{-x^2} \,\mathrm{d}y \,\mathrm{d}x.$$

Note that in order to determine the limits we should draw the region and use that to work out the new limits but as this is the same as the earlier example we refer to Figure 6.3.
Now if we evaluate the inner integral

$$\int_0^1 \int_0^x e^{-x^2} \,\mathrm{d}y \,\mathrm{d}x = \int_0^1 xe^{-x^2} \,\mathrm{d}x.$$

Observe that $-\frac{1}{2}\frac{d}{dx}(e^{-x^2}) = xe^{-x^2}$ so we have

$$\int_0^1 \int_0^x e^{-x^2} \,\mathrm{d}y \,\mathrm{d}x = -\frac{1}{2}\int_0^1 \frac{d}{dx}(e^{-x^2}) \,\mathrm{d}x = -\frac{1}{2}(e^{-1} - 1) \approx 0.316.$$

More complicated regions

In the regions we have considered so far the area R is a rectangle or a triangle however the region can get more complicated. For instance in the following example the boundary of R is not comprised of straight lines.

Example 6.8. Determine

$$\iint_R y \,\mathrm{d}x \,\mathrm{d}y$$

where $R = \{(x, y) : 0 \leq y \leq 1, x^2 + y^2 \leq 1\}$ is a unit semicircle. As always we start by drawing the region, which is done in Figure 6.4.

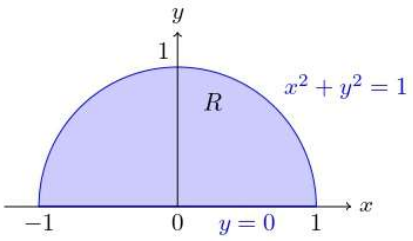


Figure 6.4

Next, we need to choose the order of integration, in this example we will show both starting with using y for the outer integral.

Solution 1: If we use y for the outer integral then we observe that y varies from 0 to 1, for fixed y we have that x runs from $-\sqrt{1-y^2}$ to $\sqrt{1-y^2}$. Therefore

$$\iint_R y\,\mathrm{d}x\,\mathrm{d}y = \int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} y\,\mathrm{d}x\,\mathrm{d}y = 2\int_0^1 y\sqrt{1-y^2}\,\mathrm{d}y.$$

Observe that $\frac{d}{dy}(1-y^2)^{3/2} = -3y(1-y^2)^{1/2}$ so we have

$$\iint_R y\,\mathrm{d}x\,\mathrm{d}y = -\frac{2}{3}\int_0^1 \frac{d}{\mathrm{d}y}(1-y^2)^{3/2}\,\mathrm{d}y = \frac{2}{3}.$$

Solution 2: Alternatively, if we use x for the outermost integral then x varies from -1 to 1, then for fixed x the variable y runs from 0 to $\sqrt{1-x^2}$. In which case

$$\begin{aligned}\iint_R y\,\mathrm{d}x\,\mathrm{d}y &= \int_{-1}^1 \int_0^{\sqrt{1-x^2}} y\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{-1}^1 \left[\frac{1}{2}y^2\right]_{y=0}^{y=\sqrt{1-x^2}} \mathrm{d}x \\ &= \frac{1}{2}\int_{-1}^1 1-x^2\,\mathrm{d}x \\ &= \frac{1}{2}\left[x - \frac{1}{3}x^3\right]_{x=-1}^{x=1} = \frac{2}{3}.\end{aligned}$$

Example 6.9. Consider the following

$$\iint_P xy\,\mathrm{d}y\,\mathrm{d}x$$

for $P = \{(x,y)|0 \le x, 0 \le y \le x, y \le 1, x \le 1+y\}$. We start by plotting P, we do this by first plotting all the constraints of P. Then by testing suitable points we judge which region is P, this has been done in Figure 6.5. We see that P consists of the union of the blue-shaded and red cross-hatched regions. We will start by determining the limits of integration using x for the outer integral. Observe that x runs from 0 to 2 however the y limits vary depending on whether $x \le 1$ (blue shaded area) or $x \ge 1$ (red shaded area). We can deal with this issue by splitting the integral and writing

$$\iint_P xy\,\mathrm{d}y\,\mathrm{d}x = \iint_{P_1} xy\,\mathrm{d}y\,\mathrm{d}x + \iint_{P_2} xy\,\mathrm{d}y\,\mathrm{d}x$$

where $P_1 = \{(x, y)|0 \leq x \leq 1, 0 \leq y \leq x\}$ is the blue shaded area and $P_2 = \{(x, y)|1 \leq x, x-1 \leq y \leq 1\}$ is the red shaded area. Now we have

$$\iint_P xy \, \mathrm{d}y \, \mathrm{d}x = \int_0^1 \int_0^x xy \, \mathrm{d}y \, \mathrm{d}x + \int_1^2 \int_{x-1}^1 xy \, \mathrm{d}y \, \mathrm{d}x.$$

Alternatively, if we swap the order of integration then we no longer need to split the integration. Indeed if we take y to be the outer integral then y runs from 0 to 1 and for fixed y we have that x runs from y to $1 + y$. This gives

$$\begin{aligned}
\iint_P xy \, \mathrm{d}y \, \mathrm{d}x &= \int_0^1 \int_y^{1+y} xy \, \mathrm{d}x \, \mathrm{d}y = \int_0^1 \left[\frac{1}{2}x^2 y\right]_{x=y}^{x=1+y} \mathrm{d}y \\
&= \frac{1}{2}\int_0^1 (1+y)^2 y - y^3 \, \mathrm{d}y \\
&= \frac{1}{2}\int_0^1 (y + 2y^2) \, \mathrm{d}y \\
&= \frac{1}{2}\left[\frac{1}{2}y^2 + \frac{2}{3}y^3\right]_{y=0}^{y=1} = \frac{7}{12}.
\end{aligned}$$

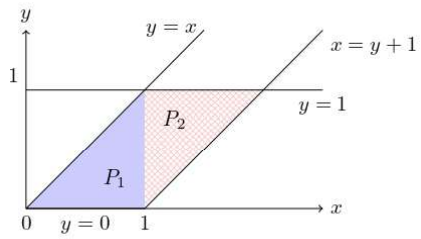


Figure 6.5

Summary of double integration

In each of our examples we have used the following steps to evaluate a double integral:

1. Draw the area of integration;
2. Decide the order of integration to use;
3. Determine the correct limits of the integration (if needed split the integral);
4. Evaluate the double integration using repeated integrals.

Useful checks to avoid errors include:

- Once you have evaluated an integral the variable of integration should no longer appear;
- If $f(x, y) \geq 0$ (for example if f is a probability density) then $\iint_R f(x, y) \, \mathrm{d}x \, \mathrm{d}y \geq 0$.